

Halloween Eye Cluster Controller

KiCad PCB Review and Layout Specification

Objective

Review the supplied KiCad project, correct obvious electrical and PCB issues, and produce a manufacturable design suitable for PCBWay assembly. The intent is refinement of an existing design, not a redesign.

Scope of Work

- Review schematic for obvious electrical mistakes (power connections, pull-ups, decoupling, missing nets, incorrect pin assignments, connector orientation, etc.).
- Run ERC and resolve errors where practical. Explain any warnings intentionally left unresolved.
- Review footprints against actual purchased components, especially ESP32-S3 module, Adafruit PCA9685 module, DFPlayer Mini modules, SSD1306 OLED header, RJ45 connectors, DIP switches, power connectors, servo headers, and PIR headers.
- Review board outline and component placement for manufacturability.
- Route all unrouted nets using good PCB practices.
- Optimize placement where necessary to simplify routing while preserving overall architecture.
- Use wide traces and copper pours for high-current 5 V servo distribution.
- Maintain solid ground return paths and appropriate decoupling.
- Separate high-current servo power from logic where practical.
- Keep ESP32 antenna free of copper beneath antenna region.
- Run DRC and resolve all errors. Minimize warnings and explain remaining ones.
- Provide updated KiCad project files.

Design Constraints

Board approximately 220 mm × 220 mm. RJ45 connectors aligned on one edge. ESP32 USB connector accessible. DFPlayer microSD cards accessible. OLED visible from board edge. Prefer simple, reliable layout over aggressive density. Do not substantially redesign the architecture without discussion.

Deliverables

1. Updated KiCad project.
2. Clean schematic.
3. Routed PCB.
4. ERC report.
5. DRC report.
6. Short summary describing changes made and any remaining concerns.
7. Gerbers are optional and should only be generated if requested.

Success Criteria

The board should pass DRC, be electrically consistent with the schematic, have clean routing suitable for PCB fabrication, and remain compatible with the specified hardware modules.